# EPM213s - Electrical Measurements

# Lab report – 4

# Exp 5: OP-amp

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Section: 5

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# OP-amp

## 1.objectives:

1. **Understand the Basic Operation of Op-Amps:**
    - **Learn how op-amps function, including the concepts of open-loop gain, input/output impedance, and common-mode rejection ratio (CMRR).**
2. **Explore Ideal vs. Practical Op-Amp Characteristics:**
    - **Compare ideal assumptions (infinite gain, infinite input impedance, zero output impedance) with real-world limitations.**
3. **Analyze and Build Basic Op-Amp Configurations:**
    - **Construct and test circuits such as:**
        - **Inverting amplifier**
        - **Non-inverting amplifier**
        - **Voltage follower (buffer)**
        - **Summing amplifier**
        - **Differential amplifier**
4. **Measure Performance Parameters:**
    - **Use laboratory instruments (oscilloscope, function generator, multimeter) to measure gain, bandwidth, offset voltage, and frequency response.**
5. **Apply Op-Amps in Signal Conditioning:**
    - **Understand how op-amps can be used to amplify, filter, or combine signals in real-world applications.**

6. **Gain Hands-On Experience with Troubleshooting:**
    - **Develop practical skills in identifying and correcting issues in op-amp circuit design and assembly.**
7. **Interpret and Analyze Experimental Results:**
    - **Compare theoretical predictions with measured results and explain discrepancies.**
8. **Develop Report Writing and Documentation Skills:**
    - **Present circuit diagrams, calculations, measured data, and analysis in a clear and professional lab report format.**

**2.procedures:**

**1. Prepare Materials and Equipment**

- **Components:**
    - **Op-amp IC (e.g., LM741 or TL081)**
    - **Resistors (e.g., 1kΩ, 10kΩ, etc.)**
    - **Breadboard and connecting wires**
- **Instruments:**
    - **DC Power supply (±15V)**
    - **Function generator**
    - **Oscilloscope or multimeter**

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**2. Understand the Theory**

- **Inverting Amplifier:**

- Vout=−(RfRin)VinV_{out} = -\left(\frac{R_f}{R_{in}}\right) V_{in}Vout=−(RinRf)Vin

- **Non-Inverting Amplifier:**
  - Vout=(1+RfR1)VinV_{out} = \left(1 + \frac{R_f}{R_1}\right) V_{in}Vout=(1+R1Rf)Vin

**Where:**

- RinR_{in}Rin or R1R_1R1 = input resistor
- RfR_fRf = feedback resistor

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## 3. Design the Circuit

- Choose resistor values to set the desired gain (e.g., gain = 10 → Rf=10kΩ,Rin=1kΩR_f = 10kΩ, R_{in} = 1kΩRf=10kΩ,Rin=1kΩ).

◆ **Inverting Amplifier:**

- Connect input signal through RinR_{in}Rin to the inverting input (-).
- Connect a feedback resistor RfR_fRf from output to inverting input.
- Connect the non-inverting input (+) to ground.

◆ **Non-Inverting Amplifier:**

- Connect input signal directly to the non-inverting input (+).
- Use voltage divider (resistors R1R_1R1 and RfR_fRf) between output and ground, with the midpoint connected to the inverting input (-).

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## 5. Apply Input Signal

- **Use function generator to apply:**
  - **A sinusoidal or DC signal to the input.**
  - **Start with small amplitude (e.g., 1 Vpp).**

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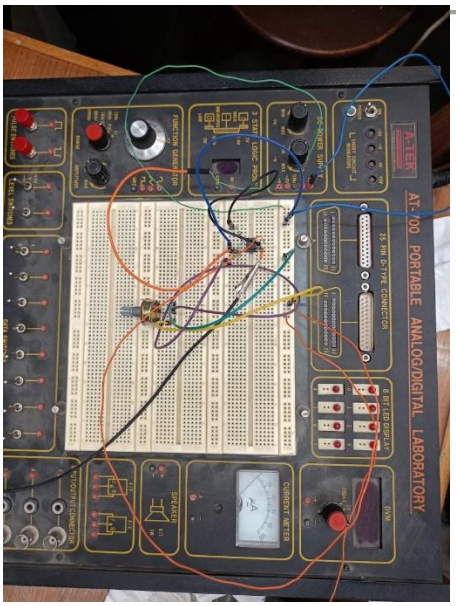
## 6. Observe and Measure Output

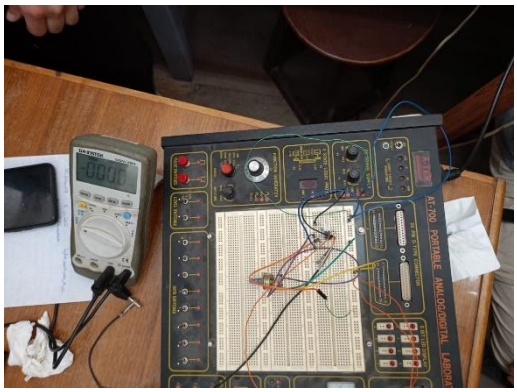
- **Use an oscilloscope or multimeter to:**
  - **Observe the amplified output waveform.**
  - **Measure gain and phase (inverting introduces 180° phase shift).**
  - **Check for signal clipping or distortion.**

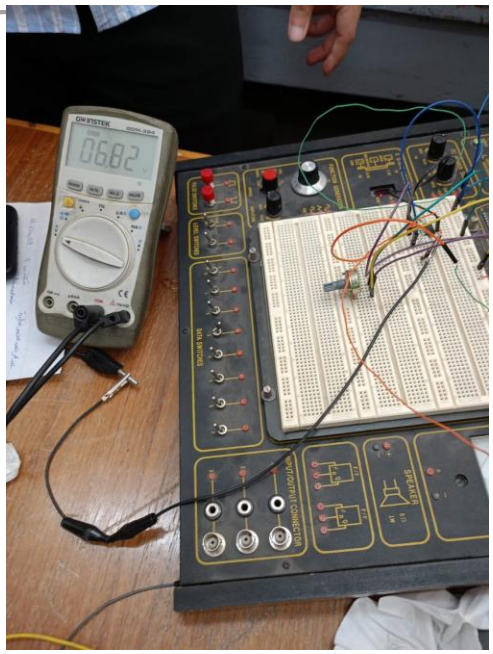
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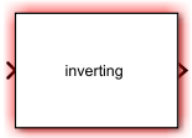
## 7. Analyze the Results

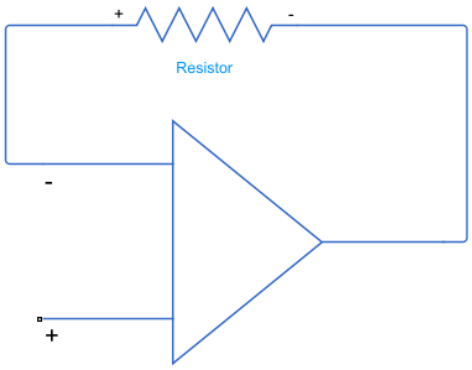
- **Calculate theoretical gain.**
- **Compare with measured gain.**
- **Explain any deviations (e.g., op-amp limitations, resistor tolerances, bandwidth effects).**

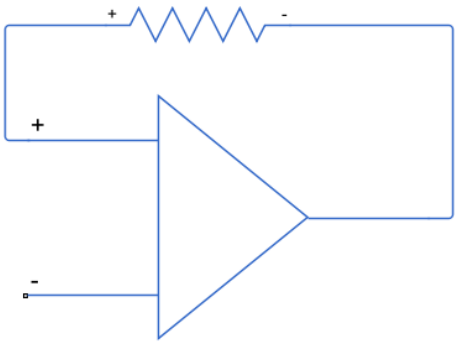





inverting


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Resistor
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